FEU INSTITUTE OF TECHNOLOGY

COURSE CODE	COURSE TITLE	UNITS / TYPE
CS0048	Structure of Programming Language	2 UNITS LEC
PREREQUISITE	Programming tools and techniques	
COURSE DESCRIPTION	This course covers specifics of writing programs in Python, this includes learning different coding conventions, program structures, and application execution. It also covers some basics of data manipulation, as well as reading and writing different types of data files.	

INSTITUTION VISION STATEMENT

FEU Institute of Technology aims to be the global standard of innovation in education, research, and nation-building by creating the next generation of changemakers that will shape the future of the country and the world.

INSTITUTION MISSION STATEMENT

FEU Institute of Technology is dedicated to provide quality, relevant, innovative, and industry-based education producing competent and principled professionals with greater sense of responsibility, social awareness and high competitiveness contributing significantly to the betterment of the society.

EDUCATIONAL PHILOSOPHY

FEU Tech's educational philosophy is rooted in transformative learning, emphasizing innovation, relevance, and social responsibility. We aim to cultivate an ecosystem of learning that shapes the next generation of leaders and innovators, driving positive change both locally and globally.

DEPARTMENT VISION STATEMENT

The Computer Science department provides specialized programs in Computer Science whose graduates are internationally recognized as innovative and specialized computing professionals.

DEPARTMENT MISSION STATEMENT

The Computer Science department is committed to provide the students and faculty with an open environment that imbues professional and personal growth through research and development.

PROGRAM EDUCATIONAL OUTCOMES

The graduates of the Bachelor of Science in Computer Science program are:

1. engaged in further professional development and have interest in or aptitude for advanced studies or trainings in computing.
2. entrepreneurs or are employed in computing industries, organizing and managing team-based projects leading to successful and sustainable computing systems solutions.
3. responsible computing professionals actively participating in community groups that make a significant impact in addressing current and future societal challenges.

PROGRAM OUTCOMES

a. Apply knowledge of computing appropriate to the discipline.
b. Analyze a complex problem and identify and define the computing requirements appropriate to its solution.
c. Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements.
d. Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal
e. Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.
f. Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions
g. Analyze the local and global impact of computing on individuals, organizations, and society.
h. Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing

professional development.

i. Apply knowledge through the use of current techniques and tools necessary for the IT profession.
j. Apply mathematical foundations, algorithmic, principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrate comprehension of the tradeoffs involved in design choices and applicable standards.
k. Apply design and development principles in the construction of software systems of varying complexity in accordance with applicable standards.
l. Apply mastery on their field of specialization

GRADUATE ATTRIBUTES

These are the skills and qualities our computing graduates will have when they finish the program.

1. Academic Education
2. Knowledge for Solving Computing Problems
3. Problem Analysis
4. Design/ Development of Solutions
5. Modern Tool Usage
6. Individual and Team Work:
7. Communication
8. Computing Professionalism and Society
9. Ethics
10. Life-long Learning

COURSE OUTCOMES AND RELATIONSHIP TO STUDENT OUTCOMES

COURSE LEARNING OUTCOMES (CLO)	INTENDED LEARNING OUTCOME	STUDENT OUTCOMES											
		a	b	c	d	e	f	g	h	i	j	k	l
CLO 1 – Python Programming Fundamentals At the end of the course, students will be able to **understand and apply fundamental Python programming concepts**, including syntax, control structures, functions, and data types, following standard coding conventions and program structures.	Module 1, 2, 3	E								E			
CLO 2 – Program Design and Implementation At the end of the course, students will be able to **design, implement, and test Python programs** using functions, object-oriented programming principles, and appropriate data structures to solve computing problems.	Module 3, 4, 5		E	E								E	
CLO 3 – Data Handling and Program Robustness At the end of the course, students will be able to **manipulate data and manage files in Python**, including reading and writing data files and implementing exception handling to ensure reliable program execution	Module 5, 6			E						E			E

WEEK / NO. OF HRS	INTENDED LEARNING OUTCOME (ILO)	DETAILED COURSE CONTENT	TEACHING AND LEARNING ACTIVITY	ASSESSMENT TASK (AT)	CLO
0 1.83 hrs.	**Module 0: Course Orientation**				
	● Orient student with the	● Introduction of subject	● Asynchronous	-	-

	course outline and topics ● Set classroom policies and guidelines	● Discussion of Syllabus ● Course Requirements ● Classroom policies	Activity ● Classroom Policy		
1 - 2 3.67 hrs	MODULE 1 – INTRODUCTION TO STRUCTURE OF PROGRAMMING LANGUAGE				
	☐ Define Python and explain its features, history, and versions ☐ Execute basic Python commands and scripts ☐ Apply basic syntax, variables, data types, and operators ●	☐ Python overview and installation ☐ Basic syntax, keywords, identifiers ☐ Variables, data types, operators ☐ User input, comments, indentation ○	☐ Lecture discussion ☐ Python installation activity ☐ Guided coding demonstrations LinkedIn Learning: Access to supplementary materials	☐ Formative quizzes ☐ Coding exercises ☐ **Summative Assessment (LQ1)**	CLO 1
3-4 3.67 hrs	MODULE 2 – CONDITIONAL STATEMENTS AND LOOPS				
	☐ Implement decision-making structures in Python ☐ Apply looping constructs to repetitive tasks ●	☐ if, if-else, if-elif-else statements ☐ for and while loops ☐ break, continue, pass ○	☐ Lecture and code walkthrough ☐ Conditional and loop exercises LinkedIn Learning: Access to supplementary materials	☐ Formative assessment ☐ Programming exercises	CLO 1
5-6 3.67 hrs	MODULE 3 – FUNCTIONS				
	☐ Define and implement Python functions ☐ Apply recursion and scope rules	☐ Function definition and arguments ☐ Recursion and anonymous functions ☐ Variable scope and lifetime ☐ Modules and packages ○	☐ Lecture discussion ☐ Function-based coding activities ●	☐ Formative assessment ☐ Summative Assessment (LQ2)	CLO 1, CLO 2
7 3 hours	**MIDTERM EXAMINATION**				
8-9 3.67 hrs	MODULE 4 – CLASSES AND OBJECTS				

	• Apply object-oriented programming concepts in Python	Classes and objects Inheritance and multiple inheritance Operator overloading	Lecture and OOP demonstrations Hands-on coding exercises	Formative assessment	CLO 2
10-11 3.67 hrs	MODULE 5 – DATA TYPES				
	• Manipulate Python built-in data types to store and process data	Numbers, strings Lists, tuples Sets, dictionaries ○	Lecture discussion Data manipulation exercises •	Formative assessment **Summative Assessment (LQ3)** •	CLO 2, CLO 3
12-13 3.67 hrs	MODULE 6 – PYTHON FILES AND EXCEPTION HANDLING				
	Perform file input/output operations Apply exception handling for program reliability	File opening, reading, writing, closing Directory operations Exception handling and user-defined exceptions ○	Lecture and demonstrations File-handling and error-handling exercises •	Formative assessment **Summative Assessment (LQ4)**	CLO 3
14 3 hrs	**Final Examination**				
Total Hours	**51 hours and 23 minutes**				

CLO	EXAMS / ASSESSMENTS
1	Formative Assessment / Recitation / Seatwork / Exercises
2	Long Quiz / Technical Summative / Laboratory Activity
3	Final Project / Final Exam / Midterm Exam

GRADING SYSTEM	
Midterm Grade	**Final Grade**
Midterm Grade (MG) = 60% (CS) + 40% (ME) **<u>Lecture: 100%</u>** • **Class Standing (CS) 60%** 1. Average of at least three long quizzes or summatives 2. Teacher's Evaluation 3. Class participation (Seatwork, Assignments, Recitations, Short Quizzes, and Formatives) 4. Lab Activities (Technical Assessment)	**Final Grade (FG) = 60% (CS) + 15% (ME) + 25% (ME)** **<u>Lecture: 100%</u>** **Class Standing (CS) 60%** 1. Average of at least three long quizzes or summatives 2. Teacher's Evaluation 3. Class participation (Seatwork, Assignments, Recitations, Short Quizzes, and Formatives)

• Lab Exercises • Machine Problems / Technicals • Practical Exam • Project ● **Midterm Exam (ME) 40%** **PASSING RAW SCORE: 70**	4. Lab Activities (Technical Assessment) ● Lab Exercises ● Machine Problems / Technicals ● Practical Exam ● Project ● **Midterm Exam (ME) 15%** ● **Final Exam (FE) 25%** **PASSING RAW SCORE: 70**

INSTRUCTIONAL MATERIALS

- Lecture Materials (PPT)/ Supplementary and Reinforcement Materials
- Video Materials / Video Courseware
- Study Guide
- Project Specification Document

TEXTBOOK/REFERENCES/ONLINE REFERENCES

Textbook / References:

- McKinney, W. (2022). *Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter* (2nd ed.). Springer.
- Kalb, I. (2022). *Object-oriented Python: Master OOP by building games and GUIs*. No Starch Press.

Online Reference

- Welcome to Python.org
- Python Archives - After Hours Programming
- Python Tutorial - Tutorialspoint
- Python Tutorials and Insights | Codementor Community
- Google's Python Class | Python Education | Google Developers
- Dive Into Python 3
- Python Tutorial | Studytonight
- Python Tutor - Visualize Python, Java, C, C++, JavaScript, TypeScript, and Ruby code execution
- Crash into Python (stephensugden.com)
- Python Tutorials – Real Python
- Full Stack Python

Online Sources:
Courseware Materials available at https://feucanvas.instructure.com

Online Journal:
EBSCO
URL:
http://search.ebscohost.com
username: feutechnologyassoc
password: aSSoc2022*

Other Journals (Available in the library):
MIS Quarterly
Journal of the ACM

Prepared by:	Approved by:	Version:
Dr. Hadji J. Tejuco Faculty-In-Charge	**Dr. Shaneth C. Ambat** Program Director, CS	Version: 1.0
Evaluated by:		Date Modified: December 2025
Mr. Abraham Magpantay, MSc Faculty Evaluator		